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Article of footwear having a closure system

Abstract

A closure system for an article of footwear includes a closure mechanism, a securing element, and a cord or a lace that is configured to be operably engaged with an upper of the footwear. The closure mechanism is configured to be actuated to adjust the footwear from a loosened configuration to a tightened configuration, and the securing element is configured to be engaged with the closure mechanism to removably secure the closure mechanism to the footwear.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of and priority to U.S. Provisional Application No. 63/418,686, filed on Oct. 24, 2022, and U.S. Provisional Application No. 63/295,984, filed on Jan. 3, 2022, which are incorporated by reference herein in their entireties.

REFERENCE REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not applicable

SEQUENCE LISTING

(2) Not applicable

BACKGROUND

1. Field of the Invention

(3) The present disclosure relates generally to an article of footwear including a fastening system or closure system.

2. Description of the Background

(4) Many conventional shoes or other articles of footwear generally comprise an upper and a sole attached to a lower end of the upper. Conventional shoes further include an internal space, i.e., a void or cavity, which is created by interior surfaces of the upper and sole, that receives a foot of a user before securing the shoe to the foot. The sole is attached to a lower surface or boundary of the upper and is positioned between the upper and the ground. As a result, the sole typically provides stability and cushioning to the user when the shoe is being worn. In some instances, the sole may include multiple components, such as an outsole, a midsole, and a top portion. The outsole may

provide traction to a bottom surface of the sole, and the midsole may be attached to an inner surface of the outsole, and may provide cushioning or added stability to the sole. For example, a sole may include a particular foam material that may increase stability at one or more desired locations along the sole, or a foam material that may reduce stress or impact energy on the foot or leg when a user is running, walking, or engaged in another activity. The sole may also include additional components, such as plates, embedded with the sole to increase the overall stiffness of the sole and reduce energy loss during use.

(5) The upper generally extends upward from the sole and defines an interior cavity that completely or partially encases a foot. In most cases, the upper extends over the instep and toe regions of the foot, and across medial and lateral sides thereof. Many articles of footwear may also include a tongue that extends across the instep region to bridge a gap between edges of medial and lateral sides of the upper, which define an opening into the cavity. The tongue may also be disposed below a lacing system and between medial and lateral sides of the upper, to allow for adjustment of shoe tightness. The tongue may further be manipulatable by a user to permit entry or exit of a foot from the internal space or cavity. In addition, the lacing system may allow a user to adjust certain dimensions of the upper or the sole, thereby allowing the upper to accommodate a wide variety of foot types having varying sizes and shapes.

(6) The upper of many shoes may comprise a wide variety of materials, which may be utilized to form the upper and chosen for use based on one or more intended uses of the shoe. The upper may also include portions comprising varying materials specific to a particular area of the upper. For example, added stability may be desirable at a front of the upper or adjacent a heel region to provide a higher degree of resistance or rigidity. In contrast, other portions of a shoe may include a soft woven textile to provide an area with stretch-resistance, flexibility, air-permeability, or moisture-wicking properties.

(7) However, in many cases, articles of footwear having uppers with an increased comfort and better fit are desired, along with improved closure mechanisms.

SUMMARY

(8) An article of footwear, as described herein, may have various configurations. The article of footwear may have an upper and a sole structure connected to the upper.

(9) In one aspect, a closure system for an article of footwear includes a closure mechanism, a securing element, and a lace that is configured to be operably engaged with an upper of the footwear. The securing element is configured to be engaged with the footwear and the closure mechanism to removably secure the closure mechanism to the footwear. The closure mechanism is configured to be rotated relative to the securing element to adjust the footwear between a loosened configuration and a tightened configuration. The lace is configured to be engaged with the closure mechanism.

(10) In some embodiments, the closure mechanism includes a collector spool that has at least one mounting tip protruding therefrom and a spool housing includes a notch that is configured to receive the at least one mounting tip. In some embodiments, the lace is passed through a plurality of eyelets formed on the upper and through at least one channel formed on the collector spool. In some embodiments, the securing element is attached to a tongue of the footwear. In some embodiments, the securing element further comprises a hook that is configured to be engaged to the tongue of the footwear. In some embodiments, the spool housing defines an interior volume and the collector spool is configured to be entirely contained within the interior volume.

(11) In some aspects, a closure system for an article of footwear includes a closure mechanism, a securing element, and a lace that is configured to be operably engaged with an upper of the footwear. The closure mechanism is configured to be actuated to adjust the footwear from a loosened configuration to a tightened configuration. The closure mechanism includes a first spool and a second spool. The first spool defines a first rotation axis and the second spool defines a second rotation axis that is disposed orthogonally relative to the first rotation axis. The first spool is

configured to be removably engaged to the second spool, and the first spool is configured to be rotated about the first rotation axis and the second rotation axis.

(12) In some embodiments, the second spool includes an upper portion, a shaft, and a lower portion. The upper portion is configured to be rotated relative to the lower portion. In some embodiments, the upper portion includes a plurality of projections, and a head cap is removably attached to the plurality of projections. In some embodiments, the first spool includes a shaft that is disposed between a first flange and a second flange, the shaft including at least one aperture to receive the lace. In some embodiments, the securing element includes a hinged mechanism for removably mounting the closure mechanism to the article of footwear. In some embodiments, the first and second flanges are configured to be rotatably engaged with a spool base. In some embodiments, the closure mechanism is removably attached to the securing element. In some embodiments, a lower portion of the closure mechanism includes a hook that is configured to be removably attached to a tongue of an article of footwear. In some embodiments, the securing element is a U-shaped member that defines a channel through which at least a portion of the lace is passed.

(13) In still another aspect, a method of assembling a closure system for an article of footwear includes the step of providing a closure mechanism comprising at least one spool that includes a shaft with at least one tab and an aperture. The method further includes the steps of providing a securing mechanism that is configured to removably secure the closure mechanism to the article of footwear and providing a lace for engagement with the closure mechanism, the lace being coupled to the shaft. The spool is rotated in a first direction to adjust the article of footwear to a tightened configuration and the spool is rotated in a second direction to adjust the article of footwear to loosened configuration.

(14) In some embodiments, the first direction is opposite the second direction. In some embodiments, the securing mechanism is disposed on an upper of the footwear.

(15) Other aspects of the article of footwear, including features and advantages thereof, will become apparent to one of ordinary skill in the art upon examination of the figures and detailed description herein. Therefore, all such aspects of the article of footwear are intended to be included in the detailed description and this summary.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a bottom and medial side of an article of footwear configured as a right shoe that includes an upper and a sole structure, according to an embodiment of the disclosure;

(2) FIG. 2 is a top view of the article of footwear of FIG. 1;

(3) FIG. 3 is a top plan view of the article of footwear of FIG. 1, with an upper removed and a user's skeletal foot structure overlaid thereon;

(4) FIG. 4 is a schematic representation of a perspective view of a lateral side of an article of footwear configured as a left shoe, according to another embodiment of the present disclosure;

(5) FIG. 5 is a schematic representation of a perspective view of a top side of a securing element;

(6) FIG. 6 is a schematic representation of an exploded view of an embodiment of a closure mechanism;

(7) FIG. 7 is a schematic representation of a perspective view of a lateral side of an article of footwear configured as a left shoe, according to another embodiment of the present disclosure;

(8) FIG. 8 is a schematic representation of an exploded view of another embodiment of a closure mechanism;

(9) FIG. 9 is a schematic representation of a perspective view of a lateral side of an article of

footwear configured as a left shoe and having a closure system, according to yet another embodiment of the present disclosure;

(10) FIG. 10 is a perspective view of a shoe upper with a closure system;

(11) FIG. 11 is a pre-assembly state or configuration of an embodiment of a closure mechanism;

(12) FIG. 12 is a pre-assembly state or configuration of an embodiment of a closure mechanism and an assembled state or configuration of one embodiment of a closure mechanism;

(13) FIG. 13 is a schematic representation of one embodiment of a closure mechanism;

(14) FIG. 14 is a schematic representation of an alternate embodiment of a closure mechanism;

(15) FIG. 15 is a schematic representation of a perspective view of a lateral side of an article of footwear configured as a right shoe having a closure system;

(16) FIG. 16 is a schematic representation of an exploded view of the closure mechanism that is configured to be removably mounted to an upper;

(17) FIG. 17 is a side view of a closure system;

(18) FIG. 18 is a perspective view of another embodiment of a lower portion of a closure system;

(19) FIG. 19 is a schematic representation of a perspective view of a lateral side of an article of footwear having a closure system;

(20) FIG. 20 is a schematic representation of an exploded view of the closure mechanism that is configured to be removably mounted to the upper;

(21) FIG. 21 is a top view of the bottom housing of a closure mechanism;

(22) FIG. 22 is a front view of an article of footwear having a closure system;

(23) FIG. 23 is a schematic representation of a perspective view of a lateral side of an article of footwear having a closure system, according to another embodiment of the present disclosure;

(24) FIG. 24 is a perspective view of one embodiment of a closure mechanism;

(25) FIG. 25 is a perspective view of one embodiment of a securing element;

(26) FIG. 26 is a perspective view of a securing element attached to a upper;

(27) FIG. 27 is an exploded view of a closure system;

(28) FIG. 28 is a perspective view of an alternate embodiment of a head cap of a closure mechanism;

(29) FIG. 29 is a perspective view of another embodiment of a head cap of a closure mechanism;

(30) FIG. 30 is a schematic representation of a perspective view of a lateral side of an article of footwear having a closure system;

(31) FIG. 31 is a perspective view of a closure mechanism in a first position;

(32) FIG. 32 is a perspective view of a closure mechanism in a second position;

(33) FIG. 33 is a perspective view of a closure mechanism in a third position;

(34) FIG. 34 is a perspective view of a closure mechanism in a fourth position;

(35) FIG. 35 is a perspective view of alternative embodiments of a head cap for a closure mechanism;

(36) FIG. 36 is a perspective view of a lateral side of an article of footwear having a closure system operatively attached to the tongue of the upper; and

(37) FIG. 37 is a perspective view of a lateral side of an article of footwear having an alternate closure system operatively attached to the tongue of the upper.

DETAILED DESCRIPTION OF THE DRAWINGS

(38) The following discussion and accompanying figures disclose various embodiments or configurations of a shoe and a sole structure. Although embodiments of a shoe or sole structure are disclosed with reference to a sports shoe, such as a running shoe, tennis shoe, basketball shoe, etc., concepts associated with embodiments of the shoe or the sole structure may be applied to a wide range of footwear and footwear styles, including cross-training shoes, football shoes, golf shoes, hiking shoes, hiking boots, ski and snowboard boots, soccer shoes and cleats, walking shoes, and track cleats, for example. Concepts of the shoe or the sole structure may also be applied to articles of footwear that are considered non-athletic, including dress shoes, sandals, loafers, slippers, and

heels. In addition to footwear, particular concepts described herein may also be applied and incorporated in other types of apparel or other athletic equipment, including helmets, padding or protective pads, shin guards, and gloves. Even further, particular concepts described herein may be incorporated in cushions, backpack straps, golf clubs, or other consumer or industrial products. Accordingly, concepts described herein may be utilized in a variety of products.

(39) The term “about,” as used herein, refers to variation in the numerical quantity that may occur, for example, through typical measuring and manufacturing procedures used for articles of footwear or other articles of manufacture that may include embodiments of the disclosure herein; through inadvertent error in these procedures; through differences in the manufacture, source, or purity of the ingredients used to make the compositions or mixtures or carry out the methods; and the like. Throughout the disclosure, the terms “about” and “approximately” refer to a range of values $\pm 5\%$ of the numeric value that the term precedes.

(40) The present disclosure is directed to an article of footwear and/or specific components of the article of footwear, such as an upper and/or a sole or sole structure. The upper may comprise a knitted component, a woven textile, and/or a non-woven textile. The knitted component may be made by knitting of yarn, the woven textile by weaving of yarn, and the non-woven textile by manufacture of a unitary non-woven web. Knitted textiles include textiles formed by way of warp knitting, weft knitting, flat knitting, circular knitting, and/or other suitable knitting operations. The knit textile may have a plain knit structure, a mesh knit structure, and/or a rib knit structure, for example. Woven textiles include, but are not limited to, textiles formed by way of any of the numerous weave forms, such as plain weave, twill weave, satin weave, dobbin weave, jacquard weave, double weaves, and/or double cloth weaves, for example. Non-woven textiles include textiles made by air-laid and/or spun-laid methods, for example. The upper may comprise a variety of materials, such as a first yarn, a second yarn, and/or a third yarn, which may have varying properties or varying visual characteristics.

(41) FIGS. 1-3 depict an exemplary embodiment of an article of footwear **100** including an upper **102** and a sole structure **104**. The upper **102** is attached to the sole structure **104** and together define an interior cavity or volume **106** (see FIGS. 2 and 3) into which a foot may be inserted. For reference, the article of footwear **100** defines a forefoot region **108**, a midfoot region **110**, and a heel region **112** (see FIGS. 2 and 3). The forefoot region **108** generally corresponds with portions of the article of footwear **100** that encase portions of the foot that includes the toes, the ball of the foot, and joints connecting the metatarsals with the toes or phalanges. The midfoot region **110** is proximate and adjoining the forefoot region **108**, and generally corresponds with portions of the article of footwear **100** that encase the arch of the foot, along with the bridge of the foot. The heel region **112** is proximate and adjoining the midfoot region **110** and generally corresponds with portions of the article of footwear **100** that encase rear portions of the foot, including the heel or calcaneus bone, the ankle, and/or the Achilles tendon.

(42) Many conventional footwear uppers are formed from multiple elements, e.g., textiles, polymer foam, polymer sheets, leather, and synthetic leather, which are joined through bonding or stitching at a seam. In some embodiments, the upper **102** of the article of footwear **100** is formed from a knitted structure or knitted components. In various embodiments, a knitted component may incorporate various types of yarn that may provide different properties to an upper. For example, one area of the upper **102** may be formed from a first type of yarn that imparts a first set of properties, and another area of the upper **102** may be formed from a second type of yarn that imparts a second set of properties. Using this configuration, properties of the upper **102** may vary throughout the upper **102** by selecting specific yarns for different areas of the upper **102**.

(43) With reference to the material(s) that comprise the upper **102**, the specific properties that a particular type of yarn will impart to an area of a knitted component may at least partially depend upon the materials that form the various filaments and fibers of the yarn. For example, cotton may provide a soft effect, biodegradability, or a natural aesthetic to a knitted material. Elastane and

stretch polyester may each provide a knitted component with a desired elasticity and recovery. Rayon may provide a high luster and moisture absorbent material, wool may provide a material with an increased moisture absorbance, nylon may be a durable material that is abrasion-resistant, and polyester may provide a hydrophobic, durable material.

(44) Other aspects of a knitted component may also be varied to affect the properties of the knitted component and provide desired attributes. For example, a yarn forming a knitted component may include monofilament yarn or multifilament yarn, or the yarn may include filaments that are each formed of two or more different materials. In addition, a knitted component may be formed using a particular knitting process to impart an area of a knitted component with particular properties. Accordingly, both the materials forming the yarn and other aspects of the yarn may be selected to impart a variety of properties to particular areas of the upper **102**.

(45) In some embodiments, an elasticity of a knit structure may be measured based on comparing a width or length of the knit structure in a first, non-stretched state to a width or length of the knit structure in a second, stretched state after the knit structure has a force applied to the knit structure in a lateral direction. In further embodiments, the upper **102** may also include additional structural elements. For example, in some embodiments, a heel plate or cover (not shown) may be provided on the heel region **112** to provide added support to a heel of a user. In some instances, other elements, e.g., plastic material, logos, trademarks, etc., may also be applied and fixed to an exterior surface using glue or a thermoforming process. In some embodiments, the properties associated with the upper **102**, e.g., a stitch type, a yarn type, or characteristics associated with different stitch types or yarn types, such as elasticity, aesthetic appearance, thickness, air permeability, or scuff-resistance, may be varied.

(46) Referring again to FIG. **1**, the sole structure **104** is connected or secured to the upper **102** and extends between a foot of a user and the ground when the article of footwear **100** is worn by the user. The sole structure **104** may include one or more components, which may include an outsole, a midsole, a heel, a vamp, and/or an insole. For example, in some embodiments, a sole structure may include an outsole that provides structural integrity to the sole structure, along with providing traction for a user, a midsole that provides a cushioning system, and an insole that provides support for an arch of a user. In addition, the insole may be a strobel board, a forefoot board, a lasting board, etc., or a combination thereof, and the insole may be provided between the upper **102** and the sole structure **104**, or the insole may be provided as part of the upper **102**.

(47) Furthermore, the insole can be positioned within the interior cavity **106** of the upper **102**, which can be in direct contact with a user's foot while an article of footwear **100** is being worn. Moreover, the upper **102** may also include a liner (not shown) that can increase comfort, for example, by reducing friction between the foot of the user and the upper **102**, the sole structure **104**, the insole, or the like, and/or by moisture wicking properties. The liner may line the entirety of the interior cavity **106** or only a portion thereof. In some embodiments, a binding (not shown) may surround an opening of the interior cavity **106** to secure the liner to the upper **102** and/or to provide an aesthetic element on the article of footwear **100**.

(48) Referring to FIGS. **2** and **3**, the article of footwear **100** also defines a lateral side **116** and a medial side **118**. When a user is wearing the shoes, the lateral side **116** corresponds with an outside-facing portion of the article of footwear **100** while the medial side **118** corresponds with an inside-facing portion of the article of footwear **100**. As such, the article of footwear **100** has opposing lateral sides **116** and medial sides **118**. The medial side **118** and the lateral side **116** adjoin one another along a longitudinal central plane or central axis **120** of the article of footwear **100**, which is coplanar with the longitudinal axis L of FIG. **1**. As will be further discussed herein, the longitudinal central plane or central axis **120** may demarcate a central, intermediate axis between the medial side **118** and the lateral side **116** of the article of footwear **100**. Put differently, the longitudinal plane or central axis **120** may extend between a rear, proximal end **122** of the article of footwear **100** and a front, distal end **124** of the article of footwear **100** and may continuously define

a middle of an insole **126**, the sole structure **104**, and/or the upper **102** of the article of footwear **100**, i.e., the longitudinal plane or central axis **120** is a straight axis extending through the rear, proximal end **122** of the heel region **112** to the front, distal end **124** of the forefoot region **108**.

(49) Unless otherwise specified, and referring to FIGS. **2** and **3**, the article of footwear **100** may be defined by the forefoot region **108**, the midfoot region **110**, and the heel region **112**. The forefoot region **108** may generally correspond with portions of the article of footwear **100** that encase portions of a foot **128** that include the toes or phalanges **130**, the ball of the foot **132**, and one or more of the joints **134** that connect the metatarsals **136** of the foot **128** with the toes or phalanges **130**. The midfoot region **110** is proximate and adjoins the forefoot region **108**. The midfoot region **110** generally corresponds with portions of the article of footwear **100** that encase an arch of a foot **128**, along with a bridge of the foot **128**. The heel region **112** is proximate to the midfoot region **110** and adjoins the midfoot region **110**. The heel region **112** generally corresponds with portions of the article of footwear **100** that encase rear portions of the foot **128**, including the heel or calcaneus bone **138**, the ankle (not shown), and/or the Achilles tendon (not shown).

(50) Still referring to FIGS. **2** and **3**, the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and the lateral side **116** are intended to define boundaries or areas of the article of footwear **100**. To that end, the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and the lateral side **116** generally characterize sections of the article of footwear **100**. Certain aspects of the disclosure may refer to portions or elements that are coextensive with one or more of the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and/or the lateral side **116**. Further, both the upper **102** and the sole structure **104** may be characterized as having portions within the forefoot region **108**, the midfoot region **110**, the heel region **112**, and/or along the medial side **118** and/or the lateral side **116**. Therefore, the upper **102** and the sole structure **104**, and/or individual portions of the upper **102** and the sole structure **104**, may include portions thereof that are disposed within the forefoot region **108**, the midfoot region **110**, the heel region **112**, and/or along the medial side **118** and/or the lateral side **116**.

(51) Still referring to FIGS. **2** and **3**, the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and the lateral side **116** are shown in detail. The forefoot region **108** extends from a toe end **140** to a widest portion **142** of the article of footwear **100**. The widest portion **142** is defined or measured along a first line **144** that is perpendicular with respect to the central axis **120** that extends from a distal portion of the toe end **140** to a distal portion of a heel end **146**, which is opposite the toe end **140**. The midfoot region **110** extends from the widest portion **142** to a thinnest portion **148** of the article of footwear **100**. The thinnest portion **148** of the article of footwear **100** is defined as the thinnest portion of the article of footwear **100** measured across a second line **150** that is perpendicular with respect to the central axis **120**. The heel region **112** extends from the thinnest portion **148** to the heel end **146** of the article of footwear **100**.

(52) It should be understood that numerous modifications may be apparent to those skilled in the art in view of the foregoing description, and individual components thereof, may be incorporated into numerous articles of footwear. Accordingly, aspects of the article of footwear **100** and components thereof, may be described with reference to general areas or portions of the article of footwear **100**, with an understanding the boundaries of the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and/or the lateral side **116** as described herein may vary between articles of footwear. However, aspects of the article of footwear **100** and individual components thereof, may also be described with reference to exact areas or portions of the article of footwear **100** and the scope of the appended claims herein may incorporate the limitations associated with these boundaries of the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and/or the lateral side **116** discussed herein.

(53) Still referring to FIGS. **2** and **3**, the medial side **118** begins at the distal, toe end **140** and bows outward along an inner side of the article of footwear **100** along the forefoot region **108** toward the

midfoot region **110**. The medial side **118** reaches the first line **144**, at which point the medial side **118** bows inward, toward the central axis **120**. The medial side **118** extends from the first line **144**, i.e., the widest portion **142**, toward the second line **150**, i.e., the thinnest portion **148**, at which point the medial side **118** enters into the midfoot region **110**, i.e., upon crossing the first line **144**. Once reaching the second line **150**, the medial side **118** bows outward, away from the longitudinal, central axis **120**, at which point the medial side **118** extends into the heel region **112**, i.e., upon crossing the second line **150**. The medial side **118** then bows outward and then inward toward the heel end **146**, and terminates at a point where the medial side **118** meets the longitudinal, central axis **120**.

(54) The lateral side **116** also begins at the distal, toe end **140** and bows outward along an outer side of the article of footwear **100** along the forefoot region **108** toward the midfoot region **110**. The lateral side **116** reaches the first line **144**, at which point the lateral side **116** bows inward, toward the longitudinal, central axis **120**. The lateral side **116** extends from the first line **144**, i.e., the widest portion **142**, toward the second line **150**, i.e., the thinnest portion **148**, at which point the lateral side **116** enters into the midfoot region **110**, i.e., upon crossing the first line **144**. Once reaching the second line **150**, the lateral side **116** bows outward, away from the longitudinal, central axis **120**, at which point the lateral side **116** extends into the heel region **112**, i.e., upon crossing the second line **150**. The lateral side **116** then bows outward and then inward toward the heel end **146**, and terminates at a point where the lateral side **116** meets the longitudinal, central axis **120**.

(55) Still referring to FIGS. **2** and **3**, the upper **102** extends along the lateral side **116** and the medial side **118**, and across the forefoot region **108**, the midfoot region **110**, and the heel region **112** to house and enclose a foot of a user. When fully assembled, the upper **102** also includes an interior surface **162** and an exterior surface **164**. The interior surface **162** faces inward and generally defines the interior cavity **106**, and the exterior surface **164** of the upper **102** faces outward and generally defines an outer perimeter or boundary of the upper **102**. The upper **102** also includes an opening **166** that is at least partially located in the heel region **112** of the article of footwear **100**, which provides access to the interior cavity **106** and through which a foot may be inserted and removed. In some embodiments, the upper **102** may also include an instep region **168** that extends from the opening **166** in the heel region **112** over an area corresponding to an instep of a foot to an area proximate the forefoot region **108**. The instep region **168** may comprise an area similar to where a tongue **170** of the present embodiment is disposed. In some embodiments, the upper **102** does not include the tongue **170**, i.e., the upper **102** is tongueless.

(56) In the illustrated embodiment, the sole structure **104** includes a midsole **172** and an outsole **174**. The outsole **174** may define a bottom end or bottom surface **176** of the sole structure **104** across the heel region **112**, the midfoot region **110**, and the forefoot region **108**. Further, the outsole **174** may be a ground-engaging portion or include a ground-engaging surface of the sole structure **104** and may be opposite of the insole thereof. As illustrated in FIG. **1**, the bottom surface **176** of the outsole **174** may include a tread pattern **178** that can include a variety of shapes and configurations. The outsole **174** may be formed from one or more materials to impart durability, wear-resistance, abrasion resistance, or traction to the sole structure **104**. In some embodiments, the outsole **174** may be formed from any kind of elastomer material, e.g., rubber, including thermoset elastomers or thermoplastic elastomers, or a thermoplastic material, e.g., thermoplastic polyurethane (TPU). In some embodiments, the outsole **174** may define a shore A hardness up to **95**. In addition, the outsole **174** may be manufactured by a process involving injection molding, vulcanization, printing layer by layer, i.e., additive manufacturing systems or methods, and the like.

(57) Still referring to FIG. **1**, the midsole **172** may be individually constructed from a thermoplastic material, such as polyurethane (PU), for example, and/or an ethylene-vinyl acetate (EVA), copolymers thereof, or a similar type of material. In other embodiments, the midsole **172** may be an EVA-Solid-Sponge (“ESS”) material, an EVA foam (e.g., PUMA® ProFoam IGNITE Foam), polyurethane, polyether, an olefin block copolymer, organosheets, a thermoplastic material (e.g., a

thermoplastic polyurethane, a thermoplastic elastomer, a thermoplastic polyolefin, etc.), or a supercritical foam. The midsole **172** may be a single polymeric material or may be a blend of materials, such as an EVA copolymer, a thermoplastic polyurethane, a polyether block amide (PEBA) copolymer, and/or an olefin block copolymer. One example of a PEBA material is PEBAX®. In some embodiments, the midsole **172** is manufactured by a process involving injection molding, vulcanization, printing layer by layer, i.e., additive manufacturing systems or methods, and the like.

(58) In embodiments where the midsole **172** is formed from a supercritical foaming process, the supercritical foam may comprise micropore foams or particle foams, such as a TPU, EVA, PEBAX®, or mixtures thereof, manufactured using a process that is performed within an autoclave, an injection molding apparatus, or any sufficiently heated/pressurized container that can process the mixing of a supercritical fluid (e.g., CO.sub.2, N.sub.2, or mixtures thereof) with a material (e.g., TPU, EVA, polyolefin elastomer, or mixtures thereof) that is preferably molten. During an exemplary process, a solution of supercritical fluid and molten material is pumped into a pressurized container, after which the pressure within the container is released, such that the molecules of the supercritical fluid rapidly convert to gas to form small pockets within the material and cause the material to expand into a foam. In further embodiments, the midsole **172** may be formed using alternative methods known in the art, including the use of an expansion press, an injection machine, a pellet expansion process, a cold foaming process, a compression molding technique, die cutting, or any combination thereof. For example, the midsole **172** may be formed using a process that involves an initial foaming step in which supercritical gas is used to foam a material and then compression molded or die cut to a particular shape.

(59) FIG. 4 depicts a schematic representation of an article of footwear **200** having a closure system **204** that includes a closure mechanism **208** and a cord **212** that is operably coupled to the upper **102** and the closure mechanism **208** to tighten and/or loosen the footwear **200**. Further, the closure system **204** includes an actuating mechanism in the form of a cable **220** having a ring end **224** that is configured to be pulled by a user's finger **228** to actuate the closure system **204** to perform one or more functions, such as, e.g., tightening, unlocking, retracting. To that end, the cable **220** is configured to be pulled away from the closure mechanism **208** to actuate one of the functions of the closure system **204**, and the cable **220** is configured to be retracted into the closure mechanism **208** to prevent entanglement, as will be described herein.

(60) Referring to FIG. 4, the upper **102** includes a base layer **232** and an overlay **236** superimposed on a portion of the base layer **232**. In particular, the overlay **236** is affixed to the base layer **232** within the forefoot region **108** and the midfoot region **110** of the upper **102**. Additionally, the overlay **236** includes a wing portion **240** that extends diagonally heel-ward, i.e., toward the heel end **146**, and laterally, i.e., toward the lateral side **116**, of the footwear **200**. The wing portion **240** is an oblong-shaped portion on which the closure mechanism **208** is located. In other embodiments, the closure mechanism **208** is attached elsewhere on the overlay **236**, the upper **102**, or the sole structure **104**. As illustrated in FIG. 4, the cord **212** is laced in an overlapping pattern along the overlay **236** and through eyelets or guides **244** located on the upper **102**. Accordingly, the guides **244** define the overlapping pattern that distributes the compression force applied by the closure system **204** for tightening the footwear **200**. In the illustrated embodiments, the guides **244** are located on the overlay **236**, although other configurations are possible. In some instances, the guides **244** may be formed by the overlay **236** or affixed to the upper **102** by the overlay **236**. In other embodiments, the guides **244** are provided on the base layer **232** of the upper **102** and separately from the overlay **236**.

(61) FIG. 5 depicts the closure mechanism **208** arranged on a securing element in the form of a panel **248** that is configured to be removably attached to the footwear **200**. In the illustrated embodiment, the panel **248** includes a first end or front **252** opposite a second end or rear **256**, and a top side **260** opposite a bottom side (not shown). In the illustrated embodiment, the closure

mechanism **208** is positioned closer to the rear **256** than to the front **252** of the panel **248**, although other positions are contemplated. The panel **248** is configured to be removably attached to the upper **102** to secure the closure mechanism **208** to the upper **102** and to conceal or hide portions of the closure mechanism **208** from exposure externally of the footwear **200**. To that end, the panel **248** may be fastened between layers of the upper **102**, such as between the overlay **236** and the base layer **232**, such that the panel **248** and the closure mechanism **208** are concealed or hidden beneath the overlay **236**. Further, the panel **248** can be fastened in various fashions, such as, e.g., frictional and/or compression forces exerted between the overlay **236** and the base layer **232** of the upper **102** when the closure system **204** is actuated to tighten the footwear **200**. In this way, the panel **248** is securely tightened to the upper **102** while the footwear **200** is tightened, and then the panel **248** is loosened for removal from the upper **102** when the footwear **200** is loosened. Additionally or alternatively, the panel **248** is attached by an adhesive, a twist-lock, by interaction with the cord **212**, by engagement with a fastener, or by insertion into a pocket (not shown) formed between or among the layers of the upper **102**, among other configurations.

(62) Referring to FIGS. **4** and **5**, the wing portion **240** of the overlay **236** can be partially affixed to the base layer **232** of the upper **102**, so as to be slipped between the base layer **232** and the overlay **236** with the rear **256** positioned beneath the wing portion **240** on the lateral side **116** of the footwear **200** and the front **252** of the panel **248** positioned closer to the medial side **118** of the footwear **200**. In this way, the rear **256** of the panel **248** is suitably shaped to cooperate with the shape of the wing portion **240** and secured by the compression and/or friction forces applied when the footwear **200** is tightened by the closure system **204**. It is also contemplated that a mounting recess (not shown) is formed on the base layer **232** to receive a portion of the closure mechanism **208**, which also assists in removably securing the closure mechanism **208** and the panel **248** to the upper **102**. Accordingly, the closure mechanism **208** is removably mounted to the upper **102** on the lateral side **116** of the footwear **200**.

(63) It is further contemplated that the closure mechanism **208** may be removably mounted to the upper **102** and without the panel **248**. In some embodiments, the closure mechanism **208** is secured to the upper **102** by a portion of the cord **212**, such that the cord **212** can be looped through the closure mechanism **208** and tightened to the upper **102** during operation of the closure system **204** to tighten the footwear **200**. In other embodiments, the closure mechanism **208** is fastened in another fashion to the upper **102** or the sole structure **104**, such as, e.g., using adhesive, hook-and-loop fasteners, zippers, snap-fit fasteners, twist-lock fasteners, and the like. Accordingly, the closure mechanism **208** is configured to be removably mounted to the footwear **200** for operation with the cord **212** to tighten and/or loosen the footwear **200**. In this way, the closure mechanism **208** is configured to be used with multiple articles of footwear, such as when a user changes shoe size or styles, and especially footwear for children at ages when shoe size changes frequently. Further, removal of the closure mechanism **208** allows the user to instead wear the footwear **200** with conventional lacing fasteners or in a loosened slip-on fashion. It is further contemplated that the closure mechanism **208** may be attached to articles of clothing, sports equipment, and luggage or storage items for tightening and/or loosening, among other uses.

(64) FIG. **6** is a schematic representation of an exploded view of the closure mechanism **208**. In the illustrated embodiment, the closure mechanism **208** includes a head cap **308** that is attached to a rotary ratchet **312** that includes a first spool **316** around which a portion of the cable **220** is configured to be collected or wound. The rotary ratchet **312** further comprises a flange **320** having a plurality of projections **324** spaced radially and configured for engagement with an arm **328**. In the illustrated embodiment, the head cap **308** and the rotary ratchet **312** are attached to a second spool **332** by a shaft **336** that defines a rotational axis **R1**. Further, the cable **220** is threaded through an aperture **340** formed in the head cap **308** so that when the cable **220** is pulled through the aperture **340**, the rotary ratchet **312** rotates about the rotational axis **R1**. Accordingly, the first spool **316** functions as a collector for the cable **220** and the second spool **332** functions as a

collector for the cord **212**.

(65) Additionally, a biasing mechanism **344** is provided in the form of a coil spring for engagement with the rotary ratchet **312** to retrieve or collect the cable **220** after being withdrawn or pulled out from the closure mechanism **208**. Further, a slip clutch **348** is fitted onto the shaft **336** for selective engagement with the second spool **332** and the rotary ratchet **312**. For example, when the slip clutch **348** is engaged with the rotary ratchet **312** and the second spool **332**, rotation of the shaft **336** is transmitted by the slip clutch **348** to the second spool **332** for collecting or winding the cord **212** about the second spool **332**. Then, when a release function of the closure mechanism **208** is actuated, the slip clutch **348** is disengaged with the rotary ratchet **312** or the second spool **332** to allow the cord **212** to be withdrawn or unwound from the second spool **332**. It is further contemplated that the cable **220** can be pulled in different directions to actuate different functions, such that a particular function is associated with a particular direction.

(66) FIG. 7 illustrates a schematic representation of another embodiment of an article of footwear **400** having a closure system **404**. In the illustrated embodiment, the closure system **404** includes a closure mechanism **408** that is operably attached to a lace **412**. The lace **412** is operably engaged with a plurality of eyelets **416** to define a winding region **420** of the upper **102**. In particular, the closure mechanism **408** is configured to be removably mounted to the upper **102** for adjusting the footwear **400** between tightened and loosened configurations. The lace **412** is passed through the eyelets **416** disposed on the upper **102** in the winding region **420** and also through a flap **424** that is attached to an overlay **236** on the upper **102**. The upper **102** also includes a pull tab **428** that is attached to the tongue **170** of the footwear **400** for allowing a user to pull or reposition the tongue **170** when tightening and loosening the footwear **400**. As illustrated in FIG. 7, the closure mechanism **408** is located within the winding region **420** and entirely exposed externally of the footwear **400** while being secured against the tongue **170**. In the illustrated embodiment, a bumper **432** is provided at the toe end **140** of the upper **102**. The bumper **432** may be provided as a thickened band of elastomeric material that extends across the toe end **140** of the upper **102** to insulate a user's toes inside the upper **102**, although other configurations are possible.

(67) FIG. 8 is a schematic representation of an exploded view of the closure mechanism **408** that is configured to be removably mounted to the upper **102**. In the illustrated embodiment, a head cap **480** is shaped to be frustoconical, i.e., a truncated cone, with indentations **484** formed on an angled sidewall **488** between a top end **492** and a lower end **496**. The indentations **484** and frustoconical shape of the head cap **480** allow for ergonomic engagement by a user when rotated to actuate the closure system **404**. In addition, the head cap **480** is attached to a collector spool **500** by a shaft **504** that defines a rotational axis R2 of the closure mechanism **408**. In the illustrated embodiment, the collector spool **500** includes an upper flange **508**, a lower flange **512**, and a passage **516** that extends transversely relative to the rotational axis R2 through the collector spool **500**. The upper and lower flanges **508**, **512** are cylindrical, although other shapes are possible.

(68) Additionally, a securing element **520** is attached to the lower flange **512** of the collector spool **500** for securing the closure mechanism **408** to the footwear **400**. In some embodiments, the securing element **520** is included as part of the closure mechanism **408**. Further, a locking interface (not shown), e.g., a plurality of intermeshed teeth, may be provided between the collector spool **500** and the securing element **520** to incrementally secure the collector spool **500** against rotation relative to the securing element **520** in a particular direction. In this particular embodiment, the securing element **520** includes a channel **530** that extends transversely relative to the rotational axis R2. The collector spool **500** can be fixedly attached to the head cap **480** to allow for simultaneous rotation therewith about the rotational axis R2. Further, the collector spool **500** may be rotatably attached to the securing element **520**, such that the securing element **520** remains fixed against rotation about the rotational axis R2 when engaged with the lace **412**. As illustrated in FIG. 8, the lace **412** includes opposing first and second ends **534**, **538** each having an aglet **542** thereon, although other configurations are possible.

(69) It is contemplated that the closure mechanism **408** can be attached to any footwear having a conventional lace similar to the footwear **400** of FIG. 7. For example, a user can pass the lace **412** through the flap **424** and less than all of the plurality of eyelets **416**, such that at least two of the eyelets **416** remain open and disengaged with the lace **412**. In some embodiments, four of the eyelets **416** and, preferably, the four of the eyelets **416** disposed closest to the pull tab **428** of the tongue **170** in the midfoot region **110**, remain disengaged with the lace **412** before attaching the closure mechanism **408** to the footwear **400**, such that the first and second ends **534**, **538** of the lace **412** are free and the footwear **400** is in a loosened configuration. With reference to FIG. 8, the user passes portions of the opposing first and second ends **534**, **538** of the lace **412** through the channel **530** of the securing element **520** in opposing directions, transversely to the rotational axis R2. In this way, the closure mechanism **408** is removably coupled to the lace **412** and, thus, to the footwear **400**.

(70) As illustrated in FIG. 7, the closure mechanism **408** is positioned in the midfoot region **110** and between the flap **424** and the pull tab **428**, although other locations are contemplated.

(71) Next, the user can pass the first and second ends **534**, **538** of the lace through the remaining eyelets **416**, or, alternatively, the user can pass the lace **412** through less than all of the eyelets **416**. Subsequently, the user passes the first and second ends **534**, **538** of the lace **412** through the passage **516** in the collector spool **500** in opposing directions, transversely to the rotational axis R2, as illustrated in FIG. 8. In some instances, the passage **516** has a smaller diameter than the channel **530**, such that when the lace **412** is received within the passage **516**, an increased frictional force prevents the lace from slipping out of the passage **516** and relative to one another. In this way, the closure mechanism **408** is configured to further secure the lace **412** against loosening when the footwear **400** is in the tightened configuration. It is contemplated that the passage **516** and the channel **530** may define identical diameters, or the passage **516** may define a larger diameter than the channel **530**. Next, the user can pull the first and second ends **534**, **538** of the lace **412** away from the collector spool **500** to tightly fasten the lace **412** and the closure mechanism **408** to the footwear **400**. In some embodiments, the user can tie a knot of any suitable kind to secure the first and second ends **534**, **538**, of the lace **412** relative to the closure mechanism **408**, although this is not necessary.

(72) Finally, the user actuates the closure mechanism **408** by rotating that head cap **480** of the closure mechanism **408** about the rotational axis R2 to wind the lace **412** about the collector spool **500**. In this way, as the first and second ends **534**, **538** of the lace **412** become wound about or collected by the collector spool **500** by rotation thereof, the footwear **400** is adjusted from the loosened configuration to a tightened configuration. Further, during rotation of the head cap **480** and the collector spool **500**, the securing element **520** is stationary about the rotational axis R2 due to the engagement with the lace **412**, and the locking interface (not shown) between the collector spool **500** and securing element **520** prevents reverse rotation of the collector spool **500** about the rotational axis R2 and relative to the securing element **520**. In this way, the closure mechanism **408** can be operated to provide incremental adjustment of the footwear **400** into the tightened configuration via engagement with the lace **412** of the footwear **400**.

(73) To adjust the footwear **400** from the tightened configuration to the loosened configuration, the user may pull the head cap **480** away from the collector spool **500** to create a spacing in which the collector spool **500** may translate axially, i.e., parallel with the rotational axis R2, away from the locking interface (not shown) on the securing element **520**. Accordingly, the collector spool **500** can freely rotate in the reverse direction about the rotational axis R2 and relative to the securing element **520**, such that the lace **412** may be unwound from the collector spool **500**. In some embodiments, the user can untie any knot that was formed to fasten the lace **412** to the collector spool **500**, and the user can locate the first and second ends **534**, **538** to unwind the lace **412** from the collector spool **500**. Further, the lace **412** is removed from the channel **530** of the securing element **520** to completely detach the closure mechanism **408** from the footwear **400**.

(74) FIG. 9 depicts a schematic representation of another embodiment of an article of footwear **600** having a closure system **604** that includes a closure mechanism **608** and a lace **612** that is operably coupled to the upper **102** and the closure mechanism **608** to tighten and/or loosen the footwear **600**. In this embodiment, elements that are shared with—i.e., that are structurally and/or functionally identical or similar to—elements present in the first embodiment (article of footwear **100**) are represented by like reference numerals. The lace **612** is operably engaged with a plurality of eyelets **616** to define a winding region **620** of the upper **102**. In particular, the closure mechanism **608** is configured to be removably mounted to the upper **102** for adjusting the footwear **600** between tightened and loosened configurations. The lace **612** is passed through the eyelets **616** disposed on the upper **102** in the winding region **620**. The eyelets **616** may be provided as openings or holes in the upper **102** alongside/adjacent the tongue **170**, and the laces **612** may be passed therethrough. Additionally or alternatively, the eyelets **616** may be provided as loops, flaps, or hooks extending from the upper and disposed along or on the tongue **170** for engagement with the laces **612**. As illustrated in FIG. 9, the closure mechanism **608** is located within the winding region **620** and entirely exposed externally of the footwear **600** while being secured against the tongue **170**. In the illustrated embodiment, the closure mechanism **608** is configured to be removably mounted to the upper **102** for adjusting the footwear **600** between tightened and/or loosened configurations.

(75) FIG. 10 is a perspective view of the upper **102** with the closure system **604**. The closure mechanism **608** includes a collector spool **700** (See FIG. 11) about which a portion of the lace **612** is configured to be wound or wrapped. Illustrative arrows **614** are provided to indicate the tightening and loosening of the laces **612** about a lateral axis L1. The closure system **604** is configured to be rotated to adjust the footwear **600** from a loosened configuration to a tightened configuration and/or from a tightened configuration to a loosened configuration. In some embodiments, the closure mechanism **608** may be rotated about a longitudinal axis L2 that is defined along an elongate direction of the closure mechanism **608**.

(76) FIG. 11 is a schematic representation of a pre-assembly state or configuration, with illustrative arrows provided to indicate an insertion path along which the lace **612** is configured to be engaged with the collector spool **700**. In the illustrated embodiment, the collector spool **700** includes a shaft **704** extending linearly between the disc-shaped flanges **701**, **702** to define a spool rotational axis R3, and a pair of tabs **724** positioned centrally on the shaft **704** between the disc-shaped flanges **701**, **702**. Each tab **724** defines a channel **730** through which a portion of the lace **612** is passed for engagement with the collector spool **700**. In some embodiments, a portion of the lace **612** and/or one end of the lace **612** is affixed within the channel **730**. It is contemplated that the user can untie any knot that was formed to fasten the lace **612** to the collector spool **700**, and then the user can locate the first and second ends **734**, **738** of the lace **612** to unwind the lace **612** from the collector spool **700**. In some embodiments, an aglet **742** of the lace **612** may include a lace lock (not shown) to affix the lace **612** with the collector spool **700** and prevent the lace **612** from being removed from the channel **730**. The first and second disc-shaped flanges **701**, **702** each include a mounting tip **744** extending axially therefrom and intersected by the rotational axis R3. In some embodiments, the first and second disc-shaped flanges **701**, **702** may be formed integrally with the shaft **704**. In operation, the laces **612** are threaded through the channels **730** of the tabs **724** and wound around the shaft **704**, which may be performed during manufacturing to provide the article of footwear **600** with pre-wound laces **612** to be adjusted by the user upon initial use. Alternatively, the footwear **600** may be provided with laces **612** that are only partially installed and require the user to complete assembly of the closure system **604** by attaching the laces **612** to the collector spool **700** prior to use. It will be appreciated that the laces **612** of the closure system **604** are configured to be accessible and removable at any time, as desired by the user or necessitated by degradation of the laces **612** during use. In some embodiments, the channels **730** may be formed transversely through the shaft **704**, as illustrated in FIG. 11. In this way, the channels **730** are configured to receive the laces **612** and allow the laces **612** to be wound about the shaft **704**. In

some embodiments, the first and the second disc-shaped flanges **701**, **702** may be rotatably attached to opposing ends of the shaft **704** to provide rotatable joint connections with the shaft **704**, such that the shaft **704** is capable of rotating about the spool rotational axis R3 relative to the first and second disc-shaped flanges **701**, **702**, which may be fixed in place and prevented from rotation. (77) In operation, the collector spool **700** is contained inside of a housing **748**, as illustrated in FIG. **12**. In the illustrated embodiment, the mounting tips **744** are received within axial notches **750** formed inside of the housing **748**. In this way, the collector spool **700** is configured to receive the laces **612**, which can be wound around the shaft **704** of the collector spool **700** upon rotation of the collector spool **700** within the housing **748**. Further, the collector spool **700** is configured to be rotated within the housing **748** about the mounting tips **744** received in the axial notches **750**, such that the collector spool **700** freely rotates within an interior cavity of the housing **748**. As the shaft **704** retrieves the laces **612**, an operative diameter of the laces **612** wound on the shaft **704** increases. When tightening the closure system **604**, there is an increase in the operative diameter of the laces **612** as greater lengths of the laces **612** are wound around the collector spool **700**. In some embodiments the axial notches **750** may be disposed about first and second disc-shaped flanges **701**, **702** and the mounting tips **744** may be disposed at the opposing ends of the housing **748** with respect to the spool rotational axis R3. A pivot joint **752** is formed by the mounting tips **744** and the axial notches **750**, which enables rotational motion of the collector spool **700** within the housing **748**. In some embodiments, the mounting tips **744** or the axial notches **750**, or some combination thereof, may be provided with surface texture, ridges, nubs, teeth, gearing, or the like to increase friction forces acting against rotation of the collector spool **700** in the housing **748**, such that the collector spool **700** is configured to resist unwinding and maintain incremental changes in tension applied to the lace **612**.

(78) FIG. **13** is a schematic representation of the housing **748** of the closure mechanism **608**. The housing **748** has a first part **754** and a second part **758** on which the axial notches **750** are formed for receiving the mounting tips **744** of the collector spool **700**. Further, a locking interface (not shown), may be provided between the collector spool **700** and the securing element **720** to incrementally secure the collector spool **700** against rotation relative to the securing element **720** in a particular direction. The mounting tip **744** and/or the axial notches **750** may function as the securing element to inhibit or selectively allow rotation. The housing **748** may include a first channel **766** and a second channel **770** disposed partially by the first part **754** and the second part **758** of the housing **748**. It is within the scope of this disclosure that mechanisms and elements disclosed in connection with the embodiments of FIGS. **7** and **8** may also be implemented with the closure system **604**.

(79) FIG. **14** is a schematic representation of another embodiment of a housing **780** of a closure mechanism **608** (see FIG. **10**), which shares similarities to the housing **748** of FIG. **13**. In the illustrated embodiment, the housing **780** comprises the first part **754** and the second part **758**. The housing **780** may include the shaft **760** comprising a first shaft part **762** that is formed adjacent to the first part **754** of the housing **780** and a second shaft part **764** that is formed adjacent to the second part **758** of the housing **780**. The first channel **766** and the second channel **770** are disposed within an upper side wall **782** of the first part **754** of the housing **780**. The second part **758** includes a lower sidewall **784** that includes a cutout **786**. The housings **748**, **780** are illustrated in FIGS. **12-14** as generally cylindrical-shaped housings. In some embodiments, the housings **748**, **780** may be provided in triangular prism shapes, cuboid shapes, polygonal shapes, rectangular shapes, irregular shapes, or any suitable shapes.

(80) FIG. **15** depicts a schematic representation of another embodiment of an article of footwear **800** having a closure system **804** that includes a closure mechanism **808** and a lace **812** that is operably coupled to the upper **102** and the closure mechanism **808** to tighten and/or loosen the footwear **800**. In this embodiment, elements that are shared with—i.e., that are structurally and/or functionally identical or similar to—elements present in the previous embodiments (See FIGS. **9-**

14) are represented by like reference numerals. The lace **812** is operably engaged with a plurality of eyelets **816** to define a winding region **820** of the upper **102**. In particular, the closure mechanism **808** is configured to be removably mounted to the upper **102** for adjusting the footwear **800** between tightened and loosened configurations. The lace **812** is passed through the eyelets **816** disposed on the upper **102** in the winding region **820**. The eyelets **816** may be provided as openings or holes in the upper alongside/adjacent the tongue **170**, and the laces **812** may be passed therethrough. Additionally or alternatively, the eyelets **816** may be provided as loops, flaps, or hooks extending from the upper and disposed along or on the tongue **170** for engagement with the laces **812**. As illustrated in FIG. 15, the closure mechanism **808** is located within the winding region **820** and entirely exposed externally of the footwear **800** while being secured against the tongue **170**. In the illustrated embodiment, the closure mechanism **808** is configured to be removably mounted to the upper **102** for adjusting the footwear **800** between tightened and loosened configurations.

(81) FIG. 16 is a schematic representation of an exploded view of the closure mechanism **808** that is configured to be removably mounted to the upper **102**. In the illustrated embodiment, a head cap **880** has indentations **884** diametrically opposed from one another and formed on an angled side wall **888** between a top end **892** and a lower end **896**. The indentations **884** are provided to receive a rotating spool **898** which rotates about a first rotational axis R4A. To retain the lace **812** along the rod **899**, a first disc-shaped flange **901** and a second disc-shaped flange **902** are disposed at the axial ends of the rod **899**. Further, the rotating spool **898** may include similar components (e.g., channels **730**) as the spool **700** illustrated in FIG. 11. The laces **812** are configured to be wound about the rod **899** that is rotated around and intersected by the first rotational axis R4A.

(82) In addition, the head cap **880** is attached to a collector spool **900** that defines a second rotational axis R4B of the closure mechanism **808**. In the illustrated embodiment, the collector spool **900** includes an upper portion **903** and a shaft **904** connecting the upper portion **903** to a lower portion **905**. In some embodiments, the rod **899** defines the first rotational axis R4A and the shaft **904** defines the second rotational axis R4B, and the first and second rotational axes R4A, R4B are disposed perpendicular to each other. The upper portion **903** includes an upper flange **908** and the lower portion **905** includes a lower flange **912**. The upper flange **908** may include a plurality of protrusions **914** extending parallel to the rotational axis R4B. In the illustrated embodiment, each protrusion of the plurality of protrusions **914** includes a beveled tip **918** for engagement with the head cap **880**. The plurality of protrusions **914** are arranged diametrically opposed from one another and formed to match the shape of the indentations **884** and the angled side walls **888** of the head cap **880**. The head cap **880** may be removably attached, e.g., snap-fit, to the collector spool **900** by engaging the plurality of protrusions **914**. In the illustrated embodiment of FIG. 17, the head cap **880** is joined with the collector spool **900** by engaging the plurality of protrusions **914** and the rotating spool **898** is disposed therebetween. The rotating spool **898** may include apertures (not shown) or fastening mechanisms (not shown) disposed along the rod **899**. Because the lace **812** is attached to the rod **899**, the lace **812** is wound around the rod **899** when the rotating spool **898** is rotated to tighten or loosen the lace **812**. As the rotating spool **898** is rotated about the first rotational axis R4A, a portion of the lace **812** is wound around, i.e., collected by, the collector spool **900**.

(83) In some embodiments, a pivot joint **916** is carried by the lower flange **912** of the lower portion **905** and extends coaxially relative to the rotational axis R4B. The upper and lower flanges **908**, **912** are cylindrical, although other shapes are contemplated. In some embodiments, the upper portion **903** and the lower portion **905** are configured to be rotated relative to one another about the pivot joint **916** and the second rotational axis R4B. As a result of this rotation, the lace **812** can be wound around the shaft **904** of the collector spool **900**. Accordingly, additional lengths of lace **812** may be wound about the rotating spool **898**, which may be used to adjust the tension of the lace **812** and to secure the rotating spool **898** in accordance with the desired tension. Referring to FIG. 16, the

lower flange **912** includes a securing element **920** extending therefrom as U-shaped member and a channel **922** is defined between the securing element **920** and the lower flange **912**. In operation, the laces **812** are threaded through the channel **922** of the securing element **920** transversely relative to the second rotational axis **R4B** to securely mount the collector spool **900** to the article of footwear **800**. In some embodiments, the laces **812** are wrapped around an exterior of the securing element **920** to securely mount the collector spool **900** to the article of footwear **800**. The rotating spool **898** is configured to be positioned atop the upper flange **908**.

(84) FIG. **18** illustrates the lower portion **905** of the collector spool **900**. In the illustrated embodiment, the lower flange **912** is generally disc-shaped, although other shapes are contemplated. The pivot joint **916** is formed, in part, by a cylindrical stem or connector **926** extending away from the lower flange **912** of the lower portion **905**. The connector **926** is configured to be attached to, e.g., snap-fit into, the shaft **904** of the upper portion **903** (see FIG. **18**) of the closure mechanism **808**, such that the upper portion **903** is capable of rotating relative to the lower portion **905**. The pivot joint **916** is disposed axially opposite of the upper flange **908** of the shaft **904**. A rectangular panel **942** is coupled to the lower flange **912** at a hinge **944** to form a hinge mechanism for securing the collector spool **900** to the article of footwear **800** (see FIG. **15**). The panel **942** has a plurality of rivets **952** that allow the lace **812** to be wrapped or wound around. In the illustrated embodiment, a locking mechanism **956** is provided to secure the panel **942** to the connector **926**, although other configurations are contemplated.

(85) FIG. **19** depicts a schematic representation of another embodiment of an article of footwear **1000** having a closure system **1004** that includes a closure mechanism **1008** and a lace **1012** that is operably coupled to the upper **102** and the closure mechanism **1008** to tighten and/or loosen the footwear **200**. The lace **1012** is operably engaged with a plurality of eyelets **1016** to define a winding region **1020** of the upper **102**. In particular, the closure mechanism **1008** is configured to be removably mounted to the upper **102** for adjusting the footwear **1000** between tightened and loosened configurations. The lace **1012** is passed through the eyelets **1016** disposed on the upper **102** in the winding region **1020**. The eyelets **1016** may be provided as openings or holes in the upper **102** alongside/adjacent the tongue **170**, and the laces **1012** may be passed therethrough. Additionally, or alternatively, the eyelets **1016** may be provided as loops, flaps, or hooks extending from the upper **102** and disposed along or on the tongue **170** for engagement with the laces **1012**. As illustrated in FIG. **19**, the closure mechanism **1008** is located within the winding region **1020** and entirely exposed externally of the footwear **400** while being secured against the tongue **170**. In the illustrated embodiment, the closure mechanism **1008** is configured to be removably mounted to the upper **102** for adjusting the footwear **1000** between tightened and loosened configurations.

(86) FIG. **20** is a schematic representation of an exploded view of the closure mechanism **1008** that is configured to be removably mounted to the upper **102**. The closure mechanism **1008** includes a spool base **1098** and a collector spool **1100**. The collector spool **1100** is removably mounted to the spool base **1098**. In the illustrated embodiment, the collector spool **1100** includes a shaft **1104** extending between first and second flanges **1108**, **1112**. The first and second flanges **1108**, **1112** are connected by the shaft **1104** and configured to rotate about a first rotational axis **R5** defined by an axial direction of the shaft **1104**. In the illustrated embodiment, the first and second flanges **1108**, **1112** are cylindrical, although other shapes are contemplated. The collector spool **1100** is configured to fit rotatably with the spool base **1098** such that the first and second flanges **1108**, **1112** are frictionally engaged with the spool base **1098** in both axial and rotational directions to provide partially inhibited rotational. Further, the spool base **1098** has a groove **1084** that allows the lace **1012** to be wound about the collector spool **1100** when rotated about the axis **R5**. The spool base **1098** may include apertures (not shown) to receive the lace **1012**. The spool base **1098** is configured to be coupled to a locking mechanism **1160** provided in the form of a lace locking disc. In the illustrated embodiment, the lace locking disc **1160** is attached to the bottom housing **1166**, although other configurations are possible. The bottom housing **1166** includes a center shaft

1168 extending from a bottom surface **1170** of the bottom housing **1166**. The center shaft **1168** includes a groove **1172** that may be used to partially join the lace locking disc **1160** with the bottom housing **1166**. The lace locking disc **1160** includes a protruding flange **1174** that extends from the bottom surface **1170** of the lace locking disc **1160** and extends partially in a radial direction about a center stem **1180**. In some embodiments, the protruding flange **1174** may be configured to serve as a locking feature for selective engagement with the lace locking disc **1160**. In operation, the laces **1012** are wound around the center stem **1180** that defines a vertical rotational axis R6 of the bottom housing **1166**. The laces **1012** are inserted through the channels **1130** of the spool base **1098** and continue winding around the lace locking disc **1160**. In some embodiments, the laces **1012** may be inserted through an aperture (not shown) within the collector spool **1100** and the lace **1012** is wrapped or wound around the collector spool **1100**. The lace locking disc **1160** is configured to be rotated about a second rotational axis R6, which is perpendicular to the first rotational axis R5. (87) FIG. 21 is a top view of the lace locking disc **1160** and the bottom housing **1166**. In use, the laces **1012** are wound around the center stem **1180** of the lace locking disc **1160** that is configured to sit within the bottom housing **1166**. The protruding flange **1174** directs the laces **1012** to wind around the center stem **1180**. In some embodiments, the lace **1012** may be wound by rotating the lace locking disc **1160** in a clockwise direction **1182** and may be unwound by rotating the lace locking disc **1160** in a counterclockwise direction.

(88) FIG. 22 is another embodiment of an article of footwear **1200** having a closure system **1204** that includes a closure mechanism **1208** in the form of a lace collector spool **1210** and a lace **1212** that is operably coupled to the upper **102** and the closure mechanism **1208** to tighten and/or loosen the footwear **1200**. In this embodiment, elements that are shared with—i.e., that are structurally and/or functionally identical or similar to—elements present in the first embodiment (article of footwear **100**) are represented by like reference numerals. The lace **1212** is operably engaged with a plurality of eyelets **1216** to define a winding region **1220** of the upper **102**. In particular, the closure mechanism **1208** is configured to be removably mounted to the upper **102** for adjusting the footwear **1200** between tightened and loosened configurations. The lace **1212** is passed through the eyelets **1216** disposed on the upper **102** in the winding region **1220**. The eyelets **1216** may be provided as openings or holes in the upper **102** alongside/adjacent the tongue **170**, and the laces **1212** may be passed therethrough. Additionally or alternatively, the eyelets **1216** may be provided as loops, flaps, or hooks extending from the upper **102** and disposed along or on the tongue **170** for engagement with the laces **1212**. As illustrated in FIG. 22, the closure mechanism **1208** is located within the winding region **1220** and entirely exposed externally of the footwear **1200**. In the illustrated embodiment, the closure mechanism **1208** is configured to be removably mounted to the upper **102** for adjusting the footwear **1200** between tightened and loosened configurations.

(89) FIG. 23 is a schematic representation of a perspective view of the lateral side **116** of an article of footwear **1200** having a closure system **1204**. The laces **1212** are wrapped or wound around the closure mechanism **1208** and knotted or tied to secure the laces **1212**. The closure mechanism **1208** is mounted onto a securing element **1222** comprising an upper portion **1224** and a lower portion **1226**. The upper portion **1224** and the lower portion **1226** form a plurality of projections **1228**. The upper portion **1224** of the plurality of projections **1228** may include a nub **1230** (see FIG. 26) and the lower portion **1226** of the plurality of projections **1228** may include a socket **1232** (see FIG. 26) to receive the nub **1230**.

(90) FIG. 24 is a perspective view of the closure mechanism **1208**. The closure mechanism **1208** includes the lace collector spool **1210** with a head cap **1234**. In the illustrated embodiment, the head cap **1234** has a conical or semi-spherical shape and is secured to a top flange **1236**. A shaft **1238** extends from the top flange **1236** opposite the head cap **1234** and axially to a bottom flange **1240**, such that the shaft **1238** connects the top flange **1236** and the bottom flange **1240**. In some embodiments, the top flange **1236**, the shaft **1238**, and the bottom flange **1240** are integrally formed as a unitary component. The bottom flange **1240** is attached to a projection **1242** which

couples to a securing element **1222** (see FIG. 23). The projection **1242** may include a plurality of ridges **1244** disposed radially about a side wall **1246** of the projection **1242**. In some embodiments, the shaft **1238** may include an aperture **1248** which receives the lace **1212**.

(91) FIG. 25 is a perspective view of the securing element **1222**. The securing element **1222** includes the plurality of projections **1228** that are configured to engage the upper **102** of the article footwear **1200** (see FIG. 23). The securing element **1222** further includes a central hole **1250** and a spring **1252** operative connected to a plunger **1254**. The plunger **1254** is configured to be biased into an extended position, such that the plunger **1254** is disposed within the central hole **1250**. In operation, the plunger **1254** is moved toward a retracted position to become displaced from the central hole **1250** and, thus, the spring **1252** becomes compressed with enough force to overcome the bias of the spring **1252**. The projection **1242** (see FIG. 24) is inserted into the central hole **1250** of the securing element **1222** and the plurality of ridges **1244** is partially engaged with the plunger **1254**. The projection **1242** may be configured to be removably attached to the securing element **1222** and may be removed as shown in FIG. 24.

(92) FIG. 26 is a perspective view of the securing element **1222** that is generally X-shaped and attached to the upper **102** and partially above the tongue **170** of the article of footwear **1200** (see FIG. 23). In some embodiments, the projections **1228** have the nubs **1230** at distal ends **1260** thereof, which are configured to be inserted through eyelets **1216** and mated with opposing distal ends **1260** of the projections **1228** that carry the sockets **1232** that are configured to receive the nubs **1230**, e.g., via a snap-fit. The securing element **1222** includes the central hole **1250** disposed at the center of the upper portion **1224** of the securing element **1222**. In some embodiments, the central hole **1250** may extend through both the upper portion **1224** and the lower portion **1226** of the securing element **1222**.

(93) FIG. 27 is an exploded view of the closure system **1204** that includes the closure mechanism **1208** and securing element **1222**. In the present disclosure, the upper portion **1224** of the securing element **1222** includes the socket **1232** and the lower portion **1226** of the securing element **1222** includes the nub **1230**. In operation, the lace **1212** wraps or winds around the shaft **1238** of the lace collector spool **1210**. As illustrated in FIG. 24, the shaft **1238** may include the aperture **1248** that partially receives the lace **1212**. The projection **1242** of the closure mechanism **1208** is configured to be connected to the central hole **1250** of the securing element **1222**.

(94) FIG. 28 is a perspective view of an alternate embodiment of a closure mechanism **1408** that includes a head cap **1410**, which is functionally similar to the head cap **1234** of FIG. 24. The head cap **1410** has a plurality of projections **1416** and indicia **1420** on an outer surface **1424** thereof.

(95) FIG. 29 is perspective view of another embodiment of a closure mechanism **1508** that includes a head cap **1510**. The head cap **1510** of the closure mechanism **1508** may be provided as a conical- or frustoconical-shaped member, although other configurations are possible. The head cap **1510** further includes a plurality of indents **1516**, an angled side wall **1520** disposed between a top end **1524** and a bottom end **1528** of the head cap **1510**.

(96) FIG. 30 is a perspective view of the lateral side **116** of an article of footwear **1900** having a closure system **1904** that includes a closure mechanism **1908** and a lace **1912** that is operably coupled to the upper **102** and the closure mechanism **1908** to tighten and/or loosen the footwear **1900**. In this embodiment, elements that are shared with—i.e., that are structurally and/or functionally identical or similar to—elements present in the first embodiment (article of footwear **100**) are represented by like reference numerals. The lace **1912** is operably engaged with a plurality of eyelets **1916** to define a winding region **1920** of the upper **102**. In particular, the closure mechanism **1908** is configured to be removably mounted to the upper **102** for adjusting the footwear **1900** between tightened and loosened configurations. The lace **1912** is passed through the eyelets **1916** disposed on the upper **102** in a winding region **1920**. The eyelets **1916** may be defined as openings or holes in the upper **102** alongside/adjacent the tongue **170**, and the laces **1912** may be passed therethrough, while alternatively the eyelets **1916** may be provided as loops, flaps, or hooks

extending from the upper **102** and disposed along or on the tongue **170** for engagement with the laces **1912**. The closure mechanism **1908** is located within the winding region **1920** and entirely exposed externally of the footwear **1900**. In the illustrated embodiment, the closure mechanism **1908** is configured to be removably mounted to the upper **102** for adjusting the footwear **1900** between tightened and loosened configurations.

(97) FIG. **31** is a perspective view of the closure mechanism **1908** in a first position **1938**. The closure mechanism **1908** includes a center button **1940**. The spool **1944** includes a shaft **1946** that extends from the center button **1940** to a base **1948**. The base **1948** may be a disc-shaped member or flange, but may be provided in other shapes. In operation, the center button **1940** is depressed while a user lifts or pulls a head cap or collar **1952** away from the center button **1940**. The collar **1952** may include indentations **1953** disposed radially around the collar **1952**, which ergonomically improve a user's grip of the collar **1952**. In some embodiments, the center button **1940** is spring-loaded and can be latched and unlatched by consecutively depressing the center button **1940**.

(98) FIG. **32** is a perspective view of the closure mechanism **1908** in a second position **1956**. In operation, the laces **1912** are fed into holes **1960** that are positioned adjacent to an upper flange **1964** positioned opposite of the base **1948**. The upper flange **1964** may be integrally formed with the center button **1940**. The shaft **1946** is positioned between the upper flange **1964** and the base **1948** of the spool **1944**. Similar to the mechanism illustrated in FIG. **16**, the upper flange **1964** may be formed integrally with the shaft **1946** and a pivot joint (not shown) may be formed at a lower end **1968** of the shaft **1946**. The pivot joint (not shown) is configured to create a rotational joint between the base **1948** and the upper flange **1964** of the spool **1944** to wind the laces **1912**.

(99) FIG. **33** is a perspective view of the closure mechanism **1908** in a third position **1972**. In use, the collar **1952** is pressed back down onto the closure mechanism **1908**. In the present disclosure, the collar **1952** may be snapped onto the upper flange **1964** (see FIG. **32**) of the spool **1944** to secure the laces **1912** within the holes **1960** disposed adjacent to the upper flange **1964** of the closure mechanism **1908**.

(100) FIG. **34** is a perspective view of the closure mechanism **1908** in a fourth position **1976**. In use, the laces **1912** are wound around the shaft **1946** of the spool **1944** and secured thereon. In the present disclosure, the upper flange **1964** of the spool **1944** is rotated in a clockwise direction to tighten the laces **1912** of the footwear **1900** and is rotated in a counterclockwise direction to loosen the laces **1912** of the footwear **1900**.

(101) FIG. **35** is a perspective view comprising another embodiment of the collar **1952**. The collar **1952** may be interchanged with collars **1982**, **1986**, **1990** of different colors, shapes and designs including with projections on the perimeter or with indentations on the perimeter.

(102) FIG. **36** is a perspective view of the lateral side **116** of the article of footwear **1900** having the closure system **1904** operatively attached to the tongue **170** of the upper **102**. The closure system **1904** includes a securing element **2010** that has a hook **2012** that attaches to a central loop **2004** of the tongue **170** of the upper **102**. In some embodiments, the hook **2012** may be connected as a living hinge, a clip, or the like.

(103) FIG. **37** is another embodiment of an article of footwear **2100** having a closure system **2104** comprising closure mechanism **2108** with a securing element **2110** provided in the form of a plug for mounting to the footwear **2100**. The closure mechanism **2108** includes similar components as illustrated in the closure mechanism **1908** illustrated in FIGS. **30-34**, such as the spool **1944**. In operation, an aperture **2114** is positioned on a bottom surface **2116** of the closure mechanism **2108**. A T-shaped protrusion **2120** extending from an inner base **2124** is fed through a hole **2128** in the tongue **170** and connects to the closure mechanism **2108**. The aperture **2114** receives the T-shaped protrusion **2120** and attaches the closure mechanism **2108** to an article of footwear **2100**.

(104) Accordingly, the closure mechanism **208**, **408**, **608**, **808**, **1008**, **1208**, **1408**, **1508**, **1908**, **2108** is configured to be used with various footwear by way of conventional laces. For example, the closure mechanism **208**, **408**, **608**, **808**, **1008**, **1208**, **1408**, **1508**, **1908**, **2108** can be provided for

use with different sizes or styles of footwear, especially footwear for children at ages when shoe size changes frequently. To that end, the closure mechanism **208, 408, 608, 808, 1008, 1208, 1408, 1508, 1908, 2108** is configured to be offered in a variety of colors. The head cap of the closure mechanism **208, 408, 608, 808, 1008, 1208, 1408, 1508, 1908, 2108** may be provided with lights (not shown), such as, e.g., LED lights, that are activated upon agitation or impact. In addition, the top end of the head caps or collars **1952, 1982, 1986, 1990** may be customized or individualized through the use of logos, lettering, symbols, numbering, or other indicia and markings. It is further contemplated that the securing element of the closure mechanism **208, 408, 608, 808, 1008, 1208, 1408, 1508, 1908, 2108** can be configured for removable attachment to the footwear **200, 400, 600, 800, 1000, 1200, 1900, 2100** without engagement with the lace, such as by engagement with the flap or the pull tab, among other configurations.

(105) It is also contemplated that the closure mechanisms **208, 408, 608, 808, 1008, 1208, 1408, 1508, 1908, 2108** may be similar to those disclosed in U.S. Pat. Nos. 5,325,613, 5,600,875, 5,606,778, 5,638,588, 5,651,198, and 5,669,116, which are all commonly assigned to Puma SE and incorporated by reference in their entirety herein. For example, it is contemplated that the closure mechanisms **208, 408, 608, 808, 1008, 1208, 1408, 1508, 1908, 2108** may include one or more aspects of such closure mechanisms to provide tightening or loosening functionality when mounted on the respective footwear **200, 400, 600, 800, 1000, 1200, 1900, 2100** of the present disclosure.

(106) In other embodiments, other configurations are possible. For example, certain features and combinations of features that are presented with respect to particular embodiments in the discussion above can be utilized in other embodiments and in other combinations, as appropriate. Further, any of the embodiments described herein may be modified to include any of the structures or methodologies disclosed in connection with other embodiments. Additionally, the present disclosure is not limited to articles of footwear of the type specifically shown. Still further, aspects of the articles of footwear of any of the embodiments disclosed herein may be modified to work with any type of footwear, apparel, or other athletic equipment.

(107) As noted previously, it will be appreciated by those skilled in the art that while the invention has been described above in connection with particular embodiments and examples, the invention is not necessarily so limited, and that numerous other embodiments, examples, uses, modifications, and departures from the embodiments, examples, and uses are intended to be encompassed by the claims attached hereto. The entire disclosure of each patent and publication cited herein is incorporated by reference, as if each such patent or publication were individually incorporated by reference herein. Various features and advantages of the invention are set forth in the following claims.

INDUSTRIAL APPLICABILITY

(108) Numerous modifications to the present invention will be apparent to those skilled in the art in view of the foregoing description. Accordingly, this description is to be construed as illustrative only and is presented for the purpose of enabling those skilled in the art to make and use the invention. The exclusive rights to all modifications which come within the scope of the appended claims are reserved.

Claims

1. A closure system for an article of footwear, comprising: a closure mechanism including a spool with at least one mounting tip protruding therefrom; a securing element including a notch of a spool housing that is configured to receive the at least one mounting tip; and a lace that is configured to be operably engaged with an upper of the footwear, wherein the securing element is configured to be engaged with the footwear and the closure mechanism to removably secure the closure mechanism to the footwear, wherein the closure mechanism is configured to be rotated relative to the securing element to adjust the footwear between a loosened configuration and a

tightened configuration, wherein the lace is configured to be engaged with the closure mechanism, wherein the spool includes a shaft and a first tab and a second tab positioned centrally on the shaft between a first disc and a second disc, wherein the first tab includes a first aperture extending therethrough and the second tab includes a second aperture extending therethrough, wherein the lace has a first end that is configured to pass through the first aperture and a second end that is configured to pass through the second aperture, and wherein the first disc and the second disc are configured to rotate with the housing.

2. The closure system of claim 1, wherein the lace is configured to be passed through a plurality of eyelets formed on the upper and through at least one channel formed on the spool.
 3. The closure system of claim 1, wherein the securing element is configured to be engaged with a tongue of the footwear.
 4. The closure system of claim 1, wherein the spool housing defines an interior volume and the spool is configured to be entirely contained within the interior volume.
 5. A method of assembling a closure system for an article of footwear, comprising the steps of: providing a closure mechanism comprising at least one spool that includes a shaft with a first tab positioned centrally on the shaft and including a first aperture that extends through the first tab, and a second tab positioned centrally on the shaft between a first disc and a second disc and including a second aperture that extends through the second tab; providing a securing mechanism that is configured to removably secure the closure mechanism to the article of footwear; and coupling a lace to the shaft for engagement with the closure mechanism, wherein a first end and a second end of the lace are configured to pass through the apertures such that the first end is configured to pass through the first aperture and the second end is configured to pass through the second aperture, wherein the spool is rotated in a first direction to adjust the article of footwear to a tightened configuration, and wherein the spool is rotated in a second direction to adjust the article of footwear to a loosened configuration.
 6. The method of claim 5, wherein the first direction is opposite of the second direction.
 7. The method of claim 6, wherein the securing mechanism is disposed on an upper of the footwear.
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